THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR CAYLEY, Sc.D., F.R.S.

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IN THE UNIVERSITY OF CAMBRIDGE.

VOL. XIII.

Pages 31–55

Papers 896 (conclusion)–903 (partial)

Scan-verified transcription

Cambridge: At the University Press.
1897.

which is

$$=\frac{1}{\lambda^2\sqrt{(A)}}(-x_2)\{D\sqrt{(A)}+B\sqrt{(E)}\};$$

and similarly the whole coefficient of dx_1 is

$$=\frac{1}{\lambda^2\sqrt{(A)}}\,(x_1)\{D\sqrt{(A)}+B\sqrt{(E)}\}.$$

Hence the terms in dx_1, dx_2 are together

$$=\frac{1}{\sqrt{(A)}}(x_1dx_2-x_2dx_1)\cdot\frac{D\sqrt{(A)}+B\sqrt{(E)}}{\lambda^2};$$

and since the terms in dy_1, dy_2 are of the like form, we have

$$dz=\left\{\frac{1}{\sqrt{(A)}}(x_1dx_2-x_2dx_1)+\frac{1}{\sqrt{(E)}}(y_1dy_2-y_2dy_1)\right\}\times\frac{D\sqrt{(A)}+B\sqrt{(E)}}{\lambda^2},$$

and combining herewith the foregoing value

$$\sqrt{(4z^2-Iz-J)}=\frac{D\sqrt{(A)}+B\sqrt{(E)}}{\lambda^2},$$

we have the required formula

$$\frac{dz}{\sqrt{(4z^3-Iz-J)}}=\left\{\frac{x_1dx_2-x_2dx_1}{\sqrt{(A)}}+\frac{y_1dy_2-y_2dy_1}{\sqrt{(E)}}\right\}.$$

As a very simple verification, suppose $a=1$, $b=c=d=e=0$; then

$$(A,\ B,\ C,\ D,\ E)=(x_1^4,\ x_1^3y_1,\ x_1^2y_1^2,\ x_1y_1^3,\ y_1^4),$$

and if $\lambda=x_1y_2-x_2y_1$, as before, then

$$z=\frac{x_1^2y_1^2}{(x_1y_2-x_2y_1)^2}=\frac{1}{\theta^2},$$

where

$$\theta=\frac{-x_1y_2}{x_1y_2-x_2y_1},\quad\text{or }\frac{1}{\theta}=\frac{x_2}{x_1}-\frac{y_2}{y_1}.$$

Also $I=0$, $J=0$, and consequently

$$\frac{dz}{\sqrt{(4z^3-Iz-J)}}=\frac{dz}{2\sqrt{(z^3)}}=\frac{d\theta}{\theta}=\frac{x_1dx_2-x_2dx_1}{x_1^2}-\frac{y_1dy_2-y_2dy_1}{y_1^2},$$

which, in virtue of the foregoing values of A, E, is

$$=\frac{x_1dx_2-x_2dx_1}{\sqrt{(A)}}-\frac{y_1dy_2-y_2dy_1}{\sqrt{(E)}}.$$

I remark that an even more simple transformation from the general quartic radical to the Weierstrassian cubic radical was obtained by Hermite; this is alluded to in my "Note sur les Covariants, &c.," *Crelle*, t. L. (1855), pp. 285–287, [135], and is given with a demonstration in my paper "Sur quelques formules pour la transformation des intégrales elliptiques," *Crelle*, t. LV.

(1858), pp. 15–24, [235], see No. IV.; viz. from the identical relation $JU^3 - IU^2H + 4H^3 = -\Phi^2$, which connects the covariants of a quartic function, it at once follows that if

$$U = (a,\ b,\ c,\ d,\ e \mid x,\ 1)^4,$$

and H is the Hessian hereof,

$$H = (ac - b^2,\ (ad - bc),\ ae + 2bd - 3c^2,\ 2(be - cd),\ ce - d^2 \mid x,\ 1)^4;$$

then writing

$$z = \frac{H}{U},$$

we have

$$\frac{dz}{\sqrt{(-4z^3 + zI - J)}} = \frac{2\,dx}{\sqrt{\{(a,\ b,\ c,\ d,\ e \mid x,\ 1)^4\}}},$$

which is the transformation in question.

897.

SUR LES RACINES D'UNE ÉQUATION ALGÉBRIQUE.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CX. (Janvier—Juin, 1890), pp. 174–176, 215–218.]

SOIT $f(u)$ une fonction rationnelle et entière avec des coefficients réels ou imaginaires, de l'ordre n; en supposant que l'équation $f'(u) = 0$, de l'ordre $n-1$, ait $n-1$ racines, je démontre que l'équation $f(u) = 0$ aura n racines. Pour cela, soit $f(u) = f(x+iy) = P + iQ$: je suppose que c dénote une quantité positive donnée, et je considère la surface $c - z = P^2 + Q^2$, en attribuant à la coordonnée z des valeurs positives; c'est seulement pour avoir des maxima au lieu de minima, et pour faciliter ainsi l'exposition, que je prends cette surface au lieu de $z = P^2 + Q^2$. On peut donner à c une valeur si grande que la courbe $c = P^2 + Q^2$ soit une courbe fermée qui ne se coupe pas, c'est-à-dire un contour simple: cela étant, on peut se figurer ce contour comme la ligne de rivage d'une île montagneuse; la valeur de z est au plus $= c$, et, en donnant à z une valeur plus petite, $= b$, on a le contour qui correspond à l'altitude b: évidemment, les contours qui correspondent à des altitudes différentes ne se coupent pas. Il s'agit de prouver que l'île a précisément n sommets, chacun de l'altitude c.

J'écris

$$\frac{dP}{dx} = X, \qquad \frac{dP}{dy} = Y;$$

donc

$$\frac{dQ}{dx} = -Y, \quad \frac{dQ}{dy} = X;$$

et, de plus,

$$\frac{dX}{dx} = a, \quad \frac{dX}{dy} = \frac{dY}{dx} = b, \quad \frac{dY}{dy} = -a.$$

et de là

$$\frac{d^2P}{dx^2} = a, \quad \frac{d^2Q}{dx^2} = -b,$$
$$\frac{d^2P}{dx\,dy} = b, \quad \frac{d^2Q}{dx\,dy} = a,$$
$$\frac{d^2P}{dy^2} = -a, \quad \frac{d^2Q}{dy^2} = b.$$

Cela étant, nous avons

$$-\tfrac{1}{2}\frac{dz}{dx} = PX - QY, \qquad -\tfrac{1}{2}\frac{dz}{dy} = PY + QX;$$

on aura un plan tangent horizontal pour $P = 0$, $Q = 0$, ou pour $X = 0$, $Y = 0$; les valeurs $P = 0$, $Q = 0$, appartiennent à la valeur c de z et correspondent à des sommets de montagne de cette altitude c; les valeurs $X = 0$, $Y = 0$ correspondent à des sommets de col. En effet, nous avons

$$-\tfrac{1}{2}\frac{d^2z}{dx^2} = Pa - Qb + X^2 + Y^2,$$

$$-\tfrac{1}{2}\frac{d^2z}{dx\,dy} = Pb + Qa,$$

$$-\tfrac{1}{2}\frac{d^2z}{dy^2} = -Pa + Qb + X^2 + Y^2;$$

donc

$$\tfrac{1}{4}\left[\frac{d^2z}{dx^2}\frac{d^2z}{dy^2}-\left(\frac{d^2z}{dx\,dy}\right)^2\right]=(X^2+Y^2)^2-(Pa-Qb)^2-(Pb+Qa)^2$$
$$=(X^2+Y^2)^2-(P^2+Q^2)(a^2+b^2),$$

valeur positive pour $P=0$, $Q=0$; négative pour $X=0$, $Y=0$.

À présent, nous avons $f'(x+iy)=X-iY$; donc, en supposant que l'équation $f'(x+iy)=0$ ait $n-1$ racines, il y aura $n-1$ systèmes de valeurs réelles de x, y qui satisfont aux équations $X=0$, $Y=0$; pour chaque système, il y aura des valeurs déterminées de P, Q et de là aussi de z; ces valeurs de z seront en général différentes. Ainsi il y aura dans l'île $n-1$ cols, dont les altitudes seront en général différentes: soient $c_1, c_2, \ldots, c_{n-1}$ ces altitudes, commençant avec la plus petite.

Pour $z=0$, nous avons un contour simple, et de même pour une valeur quelconque plus petite que c_1; mais, pour $z=c_1$, nous avons un col; le contour est une courbe, figure de 8; en donnant à z une valeur un peu plus grande, le contour se divise en deux courbes fermées, ou bien contours simples, extérieurs l'un à l'autre. Pour $z=c_2$, nous avons encore un col; l'un des contours simples s'est changé en figure de 8, et, pour une valeur un peu plus grande, le contour se divise en trois contours simples, chacun extérieur aux autres. En continuant de cette manière, on a, pour c_{n-1}, le col le plus haut; le contour est composé de $n-2$ contours simples et d'une figure de 8; et, en donnant à z une valeur un peu plus grande, on obtient un contour composé de n contours simples, chacun extérieur aux autres. Enfin, en faisant croître z, chacun des contours simples doit se réduire à un point, c'est-à-dire qu'il doit y avoir précisément n sommets de montagne; mais il n'y a pas de sommet de montagne, sinon pour la valeur $z=c$; donc il y a précisément n sommets de montagne, chacun de l'altitude c. On suppose toujours que l'équation $f'(x+iy)=0$ n'ait pas de racines égales, mais il peut bien arriver que deux ou plusieurs des valeurs $c_1, c_2, \ldots, c_n$ deviennent égales; la démonstration est très peu changée, en donnant à z une valeur un peu plus grande que celle qui correspond à l'altitude des cols d'altitude égale; le contour se divise toujours en contours simples, extérieurs chacun aux autres.

Il va sans dire que cette démonstration repose sur les mêmes principes que celles de Gauss et de Cauchy.

Je reprends la théorie des racines de l'équation $f(u)=0$; au lieu de la surface $c-z=P^2+Q^2$, il convient de considérer la surface $(c-z)^2=P^2+Q^2$, en faisant attention seulement aux valeurs de z positives et pas plus grandes que c. La théorie est très peu changée; les contours sont les mêmes qu'auparavant, mais ils appartiennent à des altitudes différentes; et, au lieu de maxima $z=c$ pour $P=0$, $Q=0$, on a des points coniques, c'est-à-dire, dans l'île montagneuse, au lieu d'un sommet arrondi de montagne, on a un cône ou un pic.

Mais avec la nouvelle surface, on construit graphiquement l'approximation de Newton: partant d'une valeur réelle ou imaginaire approximative u, on obtient la nouvelle valeur

$$u_1=u+h=u-\frac{f(u)}{f'(u)}.$$

Je représente u par le point $(x,\ y,\ z)$ de la surface $(c-z)^2=P^2+Q^2$, ou le point $(x,\ y)$ du plan des sommets $z=c$; et de même, u_1 par le point $(x_1,\ y_1,\ z_1)$ de la surface, ou $(x_1,\ y_1)$ du plan des sommets: cela étant, si, par le point $(x,\ y,\ z)$ de la surface, on mène la droite de plus grande pente (droite tangente à la surface et perpendiculaire au contour), cette droite rencontrera le plan des sommets en un point $(x_1,\ y_1)$, et l'on obtient ainsi le point $(x_1,\ y_1,\ z_1)$ de la surface, qui représente la valeur cherchée u_1. En particulier, si les coefficients de $f(u)$ sont réels, on a

$$Q=0;$$

l'équation $(c-z)^2 = P^2 + Q^2$ devient

$$(c-z)^2 = P^2,$$

c'est-à-dire

$$c - z = \pm P,$$

ou enfin

$$c - z = \pm f(x),$$

et la section verticale de l'île est formée par des parties de ces deux courbes symétriques: pour la théorie géométrique, on peut évidemment y substituer la seule courbe $c - z = f(x)$.

J'ai proposé, il y a plus de dix ans (*Amer. Math. Journ.*, t. II., 1879, [694]), le problème que je nomme "The Newton-Fourier imaginary Problem," et dans une Note (*Quart. Math. Journ.*, t. XVI., 1879, [736]), "Application of the Newton-Fourier method to an imaginary root of an equation," j'ai considéré le cas d'une équation quadratique. Pour l'équation $u^2 - 1 = 0$, on a

$$u_1 = u - \frac{u^2-1}{2u} = \tfrac{1}{2}\left(u + \frac{1}{u}\right);$$

cela donne

$$u_1 - 1 = \frac{1}{2u}(u-1)^2, \quad u_1 + 1 = \frac{1}{2u}(u+1)^2,$$

et de là

$$\frac{u_1-1}{u_1+1} = \left(\frac{u-1}{u+1}\right)^2.$$

Cette dernière équation a rapport aux deux racines $+1$ et -1, et, quoiqu'elle donne les résultats les plus élégants, cependant, en vue de la théorie générale, il vaut mieux considérer l'équation $u_1 - 1 = \frac{1}{2u}(u-1)^2$ qui se rapporte à la seule racine $+1$.

Je remarque d'abord que la formule originale $u_1 = \frac{1}{2}\left(u + \frac{1}{u}\right)$ donne

$$x_1 = \tfrac{1}{2}x\left(1 + \frac{1}{x^2+y^2}\right), \quad y_1 = \tfrac{1}{2}y\left(1 - \frac{1}{x^2+y^2}\right);$$

donc les valeurs de x et x_1 seront à la fois positives ou négatives, et ainsi l'on peut ne faire attention qu'aux valeurs positives. Cela étant, nous avons

$$x_1 + iy_1 - 1 = \frac{(x+iy-1)^2}{2(x+iy)}.$$

Désignons par A le point $(0,\ 1)$, par B le point $(0,\ -1)$, par O le point $(0,\ 0)$; et aussi par P le point $(x,\ y)$, et de même par P_1 le point $(x_1,\ y_1)$; écrivons aussi $x+iy = se^{i\psi}$, $x+iy-1 = re^{i\theta}$, $x_1 + iy_1 = r_1e^{i\theta_1}$; l'équation est

$$r_1 e^{i\theta_1} = \tfrac{1}{2}\frac{r^2e^{2i\theta}}{se^{i\psi}};$$

donc

$$r_1 = \tfrac{1}{2}\frac{r^2}{s}, \quad \left(\text{c'est-à-dire } AP_1 = \frac{AP^2}{2\,OP}\right),$$

et

$$\theta_1 = 2\theta - \psi, \quad (\text{c'est-à-dire } \widehat{AP_1x} = 2\widehat{APx} - \widehat{OPx}).$$

Je remarque que, dans la géométrie des vecteurs, la seule équation $AP_1 = \dfrac{AP^2}{2\,OP}$ dénote l'équation en $x_1 + iy_1$, $x + iy$, c'est-à-dire les deux équations que je viens de trouver.

Partant d'un point quelconque P, on obtient une suite de points $P_1, P_2, P_3, \ldots$; et, si le point P est sur l'axe des y $(x = 0)$, tous les autres points seront aussi sur l'axe de y, et l'on n'approche ni du point A ni du point B. Mais, si la coordonnée x a une valeur positive si petite que l'on veut, on arrive enfin infiniment près du point A, et l'on peut même (dans un sens qui sera expliqué plus bas, mais qui n'est pas le sens le plus naturel) dire que l'approximation est régulière. En effet, on n'a pas toujours $AP_1 < AP$, et ainsi, dans le sens le plus naturel, l'approximation n'est pas toujours régulière. Pour étudier cela, j'écris $AP_1 = AP$; cela donne $AP = 2OP$, ou, ce qui est la même chose, $x^2 + y^2 + \frac{2}{3}x = \frac{1}{3}$, c'est-à-dire que le point P sera situé sur le cercle, centre $x = -\frac{1}{3}$ et rayon $= \frac{2}{3}$; en ne faisant attention qu'aux valeurs positives de x, on a un segment compris entre l'axe des y et un arc par les points $\left(x = 0,\ y = \pm\dfrac{1}{\sqrt{3}}\right)$, $(x = \frac{1}{3},\ y = 0)$. Si le point P est sur l'arc, on aura $AP_1 = AP$; si P est en dedans du segment, alors $AP_1 > AP$; si P est en dehors du segment, $AP_1 < AP$.

Mais, en supposant P en dehors du segment, et ainsi $AP_1 < AP$, il peut bien arriver que P_1 soit en dedans du segment, et, cela étant, on aura $AP_2 > AP_1$, et l'approximation ne sera pas régulière. Mais, en considérant le cercle $x^2 + y^2 - \frac{2}{3}x = \frac{1}{3}$, lequel est le cercle, centre A et rayon $\frac{2}{3}$, qui touche le segment au point $(x = \frac{1}{3},\ y = 0)$, alors, en supposant que le point P soit en dedans de ce cercle, on aura $AP_1 < AP$, le point P_1 sera aussi en dedans du cercle, et ainsi en dehors du segment; et les points successifs $P, P_1, P_2, \ldots$ approcheront continuellement du point A; l'approximation sera dans ce cas régulière.

Il y a ainsi trois régions: le segment, le cercle $x^2 + y^2 - \frac{2}{3}x = \frac{1}{3}$ et le résidu du demi-plan; on pourrait les nommer régions *noire, blanche* et *grise* respectivement. C'est seulement pour un point P à l'intérieur de la région blanche que l'approximation est certainement régulière.

Nous venons de considérer en effet les cercles qui ont pour centre le point A; $AP_1 < AP$ veut dire que le point P est situé sur un cercle plus grand, et P_1 sur un cercle plus petit; mais, au lieu de ces cercles concentriques, considérons des cercles quelconques qui entourent le point A, sans se couper les uns les autres; et convenons de dire que c'est un bon pas quand on passe du point P sur un cercle plus grand à un point P_1 sur un cercle plus petit: avec cette convention on aura, en général, trois régions, lesquelles cependant ne seront pas les mêmes comme auparavant. En particulier, si nous considérons les cercles $AP = k\,BP$ (k une constante quelconque plus petite que l'unité), alors il n'y a pas de région noire, ou, si l'on veut, la région noire se réduit à la seule droite $y = 0$; donc il n'y a pas non plus de région grise, et le demi-plan entier est région blanche, c'est-à-dire, dans le sens que je viens d'expliquer, l'approximation est toujours régulière. En effet, c'est là la théorie à laquelle on est conduit au moyen de l'équation $\dfrac{u_1 - 1}{u_1 + 1} = \left(\dfrac{u - 1}{u + 1}\right)^2$ ci-dessus mentionnée.

En parlant de cercles, j'ai fait une restriction qui n'est nullement nécessaire; j'aurais pu parler d'ovales, de forme quelconque, qui entourent le point A sans se couper les uns les autres.

J'espère appliquer cette théorie au cas d'une équation cubique, mais les calculs sont beaucoup plus difficiles.

898.

SUR L'ÉQUATION MODULAIRE POUR LA TRANSFORMATION DE L'ORDRE 11.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CXI. (Juillet—Décembre, 1890), pp. 447–449.]

L'ÉQUATION en u, v, en y écrivant $u = x$, $v = y$, est

$$\left.\begin{array}{rl} y^{12} & \\ +\ y^{11} & (32x^3 - 22x^7) \\ +\ y^{10}\cdot & 44x^2 \\ +\ y^{9}\cdot & (88x^9 + 22x) \\ +\ y^{8}\cdot & 165x^4 \\ +\ y^{7}\cdot & 132x^9 \\ +\ y^{6}\,(- & 44x^{11} + 44x^3) \\ +\ y^{5}\cdot & -132x^8 \\ +\ y^{4}\cdot & -165x^3 \\ +\ y^{3}\,(- & 22x^{10} - 88x^2) \\ +\ y^{2}\cdot & -44x^9 \\ +\ y & (22x^{11} - 22x) \\ +\ 1\,. & -x^{12} \end{array}\right\} = 0.$$

Selon un résultat trouvé par H. J. S. Smith, pour la transformation de l'ordre p, la courbe est de l'ordre $2p$, et il y a à l'origine un point double, à l'infini deux points singuliers équivalents chacun à $\frac{1}{2}(p-1)(p-2)$ points doubles, et de plus $(p-1)(p-3)$ points doubles. Au cas $p = 11$, le nombre de ces derniers points doubles est donc $= 80$. Cela s'accorde avec l'expression

$$D = x^{12}(1-x^8)^{10}(16x^{16} - 31x^8 + 16)^2(x^{64} - 301960x^{56} + \ldots + 1)^2,$$

trouvée par M. Hermite pour le discriminant de la fonction; et l'on voit ainsi que les valeurs de x, qui correspondent aux quatre-vingts points doubles, sont données par les équations

$$16x^{16} - 31x^8 + 16 = 0,$$

$$x^{64} - 301960x^{56} + \ldots + 1 = 0.$$

Je ne considère que les seize points doubles donnés par la première équation. Cette équation donne

$$x^8 = \tfrac{1}{32}(31 + 3i\sqrt{7}), \quad x^4 = \tfrac{1}{4}(i + 3\sqrt{7}), \quad x^2 = \frac{1+i}{4\sqrt{2}}(-3 + i\sqrt{7});$$

il y a ainsi quatre points doubles, pour lesquels les valeurs de x sont

$$x = \pm\sqrt{\frac{1+i}{4\sqrt{2}}}(-3 + i\sqrt{7}), \quad x = \pm i\sqrt{\frac{1+i}{4\sqrt{2}}}(-3 + i\sqrt{7});$$

je trouve que les valeurs correspondantes de y sont $y = \dfrac{1+i}{\sqrt{2}}x$, savoir que les quatre points sont situés sur la droite $y = \dfrac{1+i}{\sqrt{2}}x$, et, en changeant successivement les signes de i et $\sqrt{2}$, on voit ainsi que les seize points sont situés, quatre à quatre, sur les droites

$$y = \frac{1+i}{\sqrt{2}}x, \quad y = \frac{1-i}{\sqrt{2}}x, \quad y = -\frac{1+i}{\sqrt{2}}x, \quad y = -\frac{1-i}{\sqrt{2}}x.$$

J'écris, pour abréger,

$$m = \frac{1+i}{\sqrt{2}}, \quad (\text{donc } m^4 = -1),$$

$$p = \tfrac{1}{4}(-3 + i\sqrt{7}),$$

donc

$$2p^2 + 3p + 2 = 0.$$

En écrivant $y = mx$ dans l'équation et en rejetant le facteur x^2, puis en écrivant $x^2 = mp$, l'équation se présente sous la forme

$$\left.\begin{array}{rl} m^{12}p^6\cdot & -32m^{12} \\ +\, m^{10}p^6\cdot & 88m^{11} \\ +\, m^9p^6\cdot & 44m^{10} - 44m^6 \\ +\, m^8p^5\cdot & -22m^{11} + 132m^7 - 22m^3 \\ +\, m^6p^5\cdot & m^{12} + 165m^8 - 165m^4 - 1 \\ +\, m^5p^5\cdot & 22m^9 - 132m^5 + 22m \\ +\, m^3p^3\cdot & 44m^6 - 44m^2 \\ +\, m^2p^3\cdot & -88m^3 \\ +\, 1\,. & 32m \end{array}\right\} = 0,$$

où les coefficients ne contiennent que les puissances m^{21}, m^{17}, m^{13}, m^9, m^5, m^1 de m et se réduisent ainsi à des multiples de m; il y a aussi un facteur numérique 8, et, en divisant par $-8m$, l'équation devient

$$4p^{10} - 11p^8 - 11p^7 + 22p^6 + 41p^5 + 22p^4 - 11p^3 - 11p^2 + 4 = 0;$$

cette équation est de la forme

$$(2p^2 + 3p + 2)^2(p^6 - 3p^5 + 2p^4 + p^3 + 2p^2 - 3p + 1) = 0.$$

La droite $y = mx$ a donc, avec la courbe, quatre intersections doubles

$$p = \tfrac{1}{4}(-3 \pm i\sqrt{7}),$$

c'est-à-dire

$$x^2 = \frac{1+i}{4\sqrt{2}}(-3 \pm i\sqrt{7});$$

on démontre sans peine que la droite n'est pas une tangente, et ces valeurs correspondent ainsi à des points doubles de la courbe, c'est-à-dire qu'il y a sur la droite $y = \dfrac{1+i}{\sqrt{2}}x$ quatre points doubles. Réciproquement, cette valeur de x^2 conduit au facteur $(16x^{16} - 31x^8 + 16)^2$ du déterminant de l'équation modulaire.

899.

SUR LES SURFACES MINIMA.

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, t. CXI. (Juillet—Décembre, 1890), pp. 953, 954.]

ON peut généraliser tant la définition que la construction de ces surfaces, en substituant pour le cercle imaginaire à l'infini une conique ou même une surface quadrique quelconque.

Je rappelle que, dans la théorie ordinaire, une surface minima est une surface telle qu'un point quelconque de la surface est situé à mi-chemin entre les deux centres de courbure, et qu'une telle surface est le lieu des points à mi-chemin entre les deux points situés respectivement sur deux courbes de longueur nulle quelconques. Or on peut rattacher la notion d'une courbe de longueur nulle à celle d'une courbe de poursuite. Pour expliquer cela, j'observe que, dans le plan, en supposant comme à l'ordinaire que le lièvre coure selon une droite et que le chien et le lièvre courent avec des vitesses uniformes, la courbe de poursuite est une courbe déterminée; mais si les vitesses varient arbitrairement, alors la définition exprime seulement que la courbe est une courbe plane. Mais, dans l'espace, si au lieu d'une droite on considère une courbe plane ou à double courbure (disons une directrice) quelconque, alors, quelles que soient les vitesses, la définition précise toujours la courbe, savoir: on a toujours pour courbe de poursuite une courbe dont chaque tangente rencontre la courbe directrice. Au lieu d'une courbe, on peut avoir une surface directrice; dans ce cas, le nom est moins applicable, néanmoins je le retiens, et je dis que la courbe de poursuite est une courbe dont chaque tangente touche la surface directrice. Nous avons, de cette manière, la définition d'une courbe de poursuite par rapport à une courbe ou surface directrice quelconque.

À présent, au lieu du cercle imaginaire à l'infini, considérons une conique quelconque, l'absolue: on établit, comme on sait, par rapport à cette conique, la notion de la perpendicularité, et ainsi les notions d'une normale et des centres de courbure ne cessent pas de subsister. On peut donc considérer une surface telle que chaque point de la surface soit l'harmonicale par rapport aux deux centres de courbure du point de rencontre de la normale avec le plan de l'absolue: on a ainsi ce que je nomme une surface quasi-minima. Il va sans dire qu'il faut modifier convenablement la notion métrique d'une aire minima pour qu'elle soit applicable à cette nouvelle surface.

Pour construire la surface, on prend par rapport à l'absolue deux courbes de poursuite quelconques, et puis sur la droite, menée par deux points quelconques de ces courbes respectivement, l'harmonicale par rapport à ces deux points du point de rencontre de la droite avec le plan de l'absolue: le lieu de ce point harmonical sera la surface quasi-minima.

Il paraît permis de substituer pour la conique absolue une surface quadrique quelconque, que je nomme aussi l'*absolue*: on a, comme on sait, les notions de la normale et des centres de courbure. Pour la surface quasi-minima, le point sur la surface sera l'un des points doubles (foyers) de l'involution formée par les deux centres de courbure et les deux points de rencontre de la normale avec l'absolue; et de même pour la construction de la surface, il faut prendre sur la droite menée par deux points quelconques des deux courbes de poursuite respectivement les points doubles (foyers) de l'involution formée par ces deux points et les deux points de rencontre de la droite avec l'absolue.

900.

JAMES JOSEPH SYLVESTER.

[(*Scientific Worthies in Nature*, XXV.); *Nature*, vol. XXXIX. (1889), pp. 217–219.]

JAMES JOSEPH SYLVESTER, born in London on September 3, 1814, is the sixth and youngest son of the late Abraham Joseph Sylvester, formerly of Liverpool[1]. He was educated at two private schools in London, and at the Royal Institution, Liverpool, whence he proceeded in due course of time to St John's College, Cambridge. In these early days he manifested considerable aptitude for mathematics, and so it was not matter for surprise that he came out in the Tripos Examination of 1837 as Second Wrangler; being incapacitated, by the fact of his Jewish origin, from taking his degree, he was not able to compete for either of the Smith's Prizes. In more enlightened times (1872), he had the degrees of B.A. and M.A., by accumulation, conferred upon him, and received therewith the honour of a Latin speech from the Public Orator. He himself says: "I am perhaps the only man in England who am a full (voting) Master of Arts for the three Universities of Dublin, Cambridge, and Oxford, having received that degree from these Universities in the order above given: from Dublin, by *ad eundem*; from Cambridge, *ob merita*; from Oxford, by decree." He is now D.C.L. of Oxford; LL.D. of Dublin and Edinburgh, and Hon. Fellow of St John's College, Cambridge. It is still open for him to receive yet higher recognition from his own alma mater[2].

Prof. Sylvester became a student of the Inner Temple, July 29, 1846, and was called to the Bar on November 22, 1850[3]. He has been Professor of Natural Philosophy at University College, London; of Mathematics at the University of Virginia, U.S.A.[4]; then ten years later Professor at the Royal Military Academy, Woolwich; and again, after a five years' interval, Professor of Mathematics at the Johns Hopkins University, Baltimore, U.S.A., from its foundation in 1877. Finally, in December 1883, he was elected Savilian Professor of Geometry at Oxford, in succession to Prof. Henry Smith[5]. His first printed paper was on Fresnel's optical theory (in the *Phil. Mag.*, 1837).

We can here only briefly allude to a communication which was accompanied by many important results: we refer to the Friday evening address (January 23, 1874) to the Royal Institution, "On Recent Discoveries in Mechanical Conversion of Motion." He says:—"It would be difficult to quote any other discovery which opens out such vast and varied horizons as this of Peaucellier's,—in one direction, descending to the wants of the workshop, the simplification of the steam-engine, the revolutionising of the mill-wright's trade, the amelioration of garden-pumps, and other domestic conveniences (the sun of science glorifies all it shines upon); and in the other, soaring to the sublimest heights of the most advanced doctrines of modern analysis, lending aid to, and throwing light from a totally unexpected quarter on the researches of such men as Abel, Riemann, Clebsch, Grassmann, and Cayley. Its head towers above the clouds, while its feet plunge into the bowels of the earth."

The only works that Prof. Sylvester has published, we believe, are: (1) "A Probationary Lecture on Geometry, delivered before the Gresham Committee and the Members of the Common Council of the City of London, December 4, 1854," a slight thing which had to be written and

[1] Foster's *Hand-book of Men at the Bar.*

[2] [** The honorary degree Sc.D. was conferred upon him by Cambridge in 1890.]

[3] Foster, l.c.

[4] The late Prof. Key, of University College and School, was the first occupant of the Chair, founded by Mr Jefferson, once President of the United States, in 1824.

[5] He commences his Oxford lecture (*Nature*, vol. XXXIII. p. 222), of date December 12, 1885, with the words: "It is now two years and seven days since a message by the Atlantic cable containing the single word 'elected' reached me in Baltimore informing me that I had been appointed Savilian Professor of Geometry in Oxford, so that for three weeks I was in the unique position of filling the post and drawing the pay of Professor of Mathematics in each of two Universities."

delivered at a few hours' notice; (2) "Laws of Verse," in 1870; (3) several short poems, sonnets, and translations, which have appeared in our columns and elsewhere.

Our notice would be incomplete without some record of the honours that have been conferred upon Dr Sylvester. He was elected a Fellow of the Royal Society on April 25, 1839; has received a Royal Medal (1860) and the Copley Medal (1880), this latter rarely awarded, we believe, to a pure mathematician. On this last occasion, Mr Spottiswoode accompanied the presentation with the words, "His extensive and profound researches in pure mathematics, especially his contributions to the theory of invariants and covariants, to the theory of numbers and to modern geometry, may be regarded as fully establishing Mr Sylvester's claim to the award." He is a Fellow of New College, Oxford; Foreign Associate of the United States National Academy of Sciences; Foreign Member of the Royal Academy of Sciences, Göttingen, of the Royal Academy of Sciences of Naples, and of the Academy of Sciences of Boston; Corresponding Member of the Institute of France, of the Imperial Academy of Science of St Petersburg, of the Royal Academy of Science of Berlin, of the Lyncei of Rome, of the Istituto Lombardo, and of the Société Philomathique. He has been long connected with the editorial staff of the *Quarterly Journal of Mathematics* (under one or another of its titles), and was the first editor of, and is a considerable contributor to, the *American Journal of Mathematics*; and he was at one time Examiner in Mathematics and Natural Philosophy in the University of London. He was not an original member of the London Mathematical Society (founded January 16, 1865), but was elected a member on June 19, 1865, Vice-President on January 15, 1866, and succeeded Prof. De Morgan as the second President on November 8, 1866. The Society showed its recognition of his great services to them and to mathematical science generally by awarding him its De Morgan Gold Medal in November 1887. Wherever Dr Sylvester goes, there is sure to be mathematical activity; and the latest proof of this is the formation, during the last term at Oxford, of a Mathematical Society, which promises, we hear without surprise, to do much for the advancement of mathematical science there.

The writings of Sylvester date from the year 1837; the number of them in the Royal Society Index up to the year 1863 is 112, in the next ten years 38, and in the volume for the next ten years 81, making 231 for the years 1837 to 1883: the number of more recent papers is also considerable. They relate chiefly to finite analysis, and cover by their subjects a large part of it: algebra, determinants, elimination, the theory of equations, partitions, tactic, the theory of forms, matrices, reciprocants, the Hamiltonian numbers, &c.; analytical and pure geometry occupy a less prominent position; and mechanics, optics, and astronomy are not absent. A leading feature is the power which is shown of originating a theory or of developing it from a small beginning; there is a breadth of treatment and determination to make the most of a subject, an appreciation of its capabilities, and real enjoyment of it. There is not unfrequently an adornment or enthusiasm of language which one admires, or is amused with: we have a motto from Milton, or Shakespeare; a memoir is a trilogy divided into three parts, each of which has its action complete within itself, but the same general cycle of ideas pervades all three, and weaves them into a sort of complex unity; the apology for an unsymmetrical solution is—symmetry, like the grace of an eastern robe, has not unfrequently to be purchased at the expense of some sacrifice of freedom and rapidity of action; and, he remarks, may not music be described as the mathematic of sense, mathematic as the music of the reason? the soul of each the same! &c. It is to be mentioned that there is always a generous and cordial recognition of the merit of others, his fellow-workers in the science.

It would be in the case of any first-rate mathematician—and certainly as much so in this as in any other case—extremely interesting to go carefully through the whole of a long list of memoirs, tracing out as well their connexion with each other, and the several leading ideas on which they depend, as also their influence on the development of the theories to which they relate; but for doing this properly, or at all, space and time, and a great amount of labour, are required. Short of doing so, one can only notice particular theorems—and there are, in the case of Sylvester, many of these, "beautiful exceedingly," which, for their own sakes, one is tempted to refer to—or one can give titles, which, to those familiar with the memoirs themselves, will recall the rich stores of investigation and theory contained therein.

A considerable number of papers, including some of the earliest ones, relate to the question of the reality of the roots of a numerical equation: in the several connexions thereof with Sturm's theorem, Newton's rule for the number of imaginary roots, and the theory of invariants. Sylvester obtained for the Sturmian functions, divested of square factors, or say for the reduced Sturmian functions, singularly elegant expressions in terms of the roots, viz. these were

$$f_2(x) = \Sigma(a-b)^2(x-c)(x-d)\ldots,\quad f_3(x) = \Sigma(a-b)^2(a-c)^2(b-c)^2(x-d)\ldots,\ \&c.;$$

but not only this: applying the Sturmian process of the greatest common measure (not to $f(x)$, $f'(x)$, but instead) to two independent functions $f(x)$, $\phi(x)$, he obtained for the several resulting functions expressions involving products of differences between the roots of the one and the other equation, $f(x)=0$, $\phi(x)=0$; the question then arose, what is the meaning of these functions? The answer is given by his theory of *intercalations*: they are signaletic functions, indicating in what manner (when the real roots of the two equations are arranged in order of magnitude) the roots of the one equation are intercalated among those of the other. The investigations in regard to Newton's rule (not previously demonstrated) are very important and valuable: the principle of Sturm's demonstration is applied to this wholly different question: viz. x is made to vary continuously, and the consequent gain or loss of changes of sign is inquired into. The third question is that of the determination of the character of the roots of a quintic equation by means of invariants. In connexion with it, we have the noteworthy idea of facultative points; viz. treating as the coordinates of a point in n-dimensional space those functions of the coefficients which serve as criteria for the reality of the roots, a point is facultative or non-facultative according as there is, or is not, corresponding thereto any equation with real coefficients: the determination of the characters of the roots depends (and, it would seem, depends only) on the bounding surface or surfaces of the facultative regions, and on a surface depending on the discriminant. Relating to these theories there are two elaborate memoirs, "On the Syzygetic Relations &c." and "Algebraical Researches &c." in the *Philosophical Transactions* for the years 1853 and 1864 respectively; but as regards Newton's rule later papers must also be consulted.

In the years 1851–54, we have various papers on homogeneous functions, the calculus of forms, &c. (*Camb. and Dub. Math. Journal*, vols. VI. to IX.), and the separate work "On Canonical Forms" (London, 1851). These contain crowds of ideas, embodied in the new words, cogredient, contragredient, concomitant, covariant, contravariant, invariant, emanant, combinant, commutant, canonical form, plexus, &c., ranging over and vastly extending the then so-called theories of linear transformations and hyperdeterminants. In particular, we have the introduction into the theory of the very important idea of continuous or infinitesimal variation: say that a function, which (whatever are the values of the parameters on which it depends) is invariant for an infinitesimal change of the parameters, is absolutely invariant.

There is, in 1844, in the *Philosophical Magazine*, a valuable paper, "Elementary Researches in the Analysis of Combinatorial Aggregation," and the titles of two other papers, 1865 and 1866, may be mentioned: "Astronomical Prolusions; commencing with the instantaneous proof of Lambert's and Euler's theorems, and modulating through the construction of the orbit of a heavenly body from two heliocentric distances, the subtended chord, and the periodic time, and the focal theory of Cartesian ovals, into a discussion of motion in a circle and its relation to planetary motion"; and the sequel thereto, "Note on the periodic changes of orbit under certain circumstances of a particle acted upon by a central force, and on vectorial coordinates, &c., together with a new theory of the analogues of the Cartesian ovals in space."

Many of the later papers are published in the *American Mathematical Journal*, founded, in 1878, under the auspices of the Johns Hopkins University, and for the first six volumes of which Sylvester was editor-in-chief. We have, in vol. I., a somewhat speculative paper entitled "An application of the new atomic theory to the graphical representation of the invariants and covariants of binary quantics," followed by appendices and notes relating to various special points of the theory; and in the same and subsequent volumes various memoirs on binary and ternary quantics, including papers (by himself, with the aid of Franklin) containing tables of the numerical generating functions for binary quantics of the first ten orders, and for simultaneous binary quantics of the first four orders, &c. The memoir (vols. II. and III.) on "Ternary cubic-form equations" is connected with some early papers relating to the theory of numbers. We have in it the theory of residuation on a cubic curve, and the beautiful chain-rule of rational derivation; viz. from an arbitrary point 1 on the curve it is possible to derive the singly infinite series of points $(1, 2, 4, 5, \ldots, 3j \pm 1)$ such that the chord through any two points, m, n, again meets the curve in a point $m + n$, $m \sim n$ (whichever number is not divisible by 3) of the series; moreover, the coordinates of any point m are rational and integral functions of the degree m^2 of those of the point 1.

There is in vol. V. the memoir, "A Constructive Theory of Partitions arranged in three acts, an Interact in two parts, and an Exodion," and in vol. VI. we have "Lectures on the Principles of Universal Algebra," (referring to a course of lectures on multinomial quantity, in the year 1881). The memoir is incomplete, but the general theories of nullity and vacuity, and of the corpus formed by two independent matrices of the same order, are sketched out; and there are, in the *Comptes rendus* of the French Academy, later papers containing developments of various points of the theory,—the conception of "nivellators" may be referred to.

The last-mentioned paper in the *American Mathematical Journal* was published subsequently to Sylvester's return to England on his appointment as Savilian Professor of Mathematics at Oxford. In December 1886, he gave there a public lecture containing an outline of his new theory of reciprocants (reported in *Nature*, January 7, 1887), and the lectures since delivered are published under the title, "Lectures on the Theory of Reciprocants" (reported by J. Hammond), same Journal, vols. VIII. to X.; thirty-three lectures actually delivered, entire or in abstract, in the course of three terms, to a class in the University, with a concluding so-called lecture 34, which is due to Hammond. The subject, as is well known, is that of the functions of a dependent variable, y, and its differential coefficients, y', y'', ..., in regard to x (or, rather, the functions of y', y'', ...), which remain unaltered by the interchange of the variables x and y: this is a less stringent condition than that imposed by Halphen ("Thèse," 1878) on his differential invariants, and the theory is accordingly a more extensive one. A passage may be quoted:—"One is surprised to reflect on the change which is come over Algebra in the last quarter of a century. It is now possible to enlarge to an almost unlimited extent on any branch of it. These thirty lectures, embracing only a fragment of the theory of reciprocants, might be compared to an unfinished epic in thirty cantos. Does it not seem as if Algebra had attained to the dignity of a fine art, in which the workman has a free hand to develop his conceptions, as in a musical theme or a subject for painting? Formerly, it consisted in detached theorems, but nowadays it has reached a point in which every properly-developed algebraical composition, like a skilful landscape, is expected to suggest the notion of an infinite distance lying beyond the limits of the canvas." And, indeed, the theory has already spread itself out far and wide, not only in these lectures by its founder, but in various papers by auditors of them, and others,—Elliott, Hammond, Leudesdorf, Rogers, Macmahon, Berry, Forsyth.

Sylvester's latest important investigations relate to the Hamiltonian numbers: there is a memoir, *Crelle*, t. C. (1887), and, by Sylvester and Hammond jointly, two memoirs in the *Philosophical Transactions*. The subject is that of the series of numbers 2, 3, 5, 11, 47, 923, calculated thus far by Sir W. R. Hamilton in his well-known Report to the British Association, on Jerrard's method. A formula for the independent calculation of any term of the series was obtained by Sylvester, but the remarkable law by means of a generating function was discovered by Hammond, viz. $E_0, E_1, E_2, \ldots$, being the series 3, 4, 6, ... of the foregoing numbers, each increased by unity; then these are calculated by the formula

$$(1-t)^{E_0} \times (1-t^2)^{E_1} \times (1-t^4)^{E_2} \times \ldots = 1 - 2t,$$

equating the powers of t on the two sides respectively: observe the paradox, $t = \frac{1}{2}$, then the formula gives $0 =$ sum of a series of positive powers of $\frac{1}{2}$.

Enough has been said to call to mind some of Sylvester's achievements in mathematical science. Nothing further has been attempted in the foregoing very imperfect sketch.

901.

NOTE ON THE SUMS OF TWO SERIES.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 29–31.]

I CONSIDER the two series

$$S_1 = \frac{1}{1+e^{\pi a}} + \frac{1}{3(1+e^{3\pi a})} + \frac{1}{5(1+e^{5\pi a})} + \cdots$$

and

$$S = \frac{1}{1+e^{\pi a}} + \frac{1}{2(1+e^{2\pi a})} + \frac{1}{3(1+e^{3\pi a})} + \cdots\,,$$

where a is real, positive, and indefinitely small; these would at first sight appear to be equal to each other, but this is not in fact the case.

Taking first the series S_1, putting therein $\pi a = 2x$, this is

$$2S_1 = \frac{1}{1+e^{2x}} + \frac{1}{3(1+e^{6x})} + \frac{1}{5(1+e^{10x})} + \cdots\,.$$

Now we have, (Legendre, *Théorie des Fonctions Elliptiques*, t. II. p. 438),

$$\frac{1}{1+y} + \frac{1}{3(1+y^3)} + \frac{1}{5(1+y^5)} + \cdots = \tfrac{1}{2}\log\Gamma(y+1),$$

where C is Euler's constant, $= \cdot577\ldots$; and if y be real, positive, and very large, then

$$\Gamma(y+1) = e^{(y+\frac{1}{2})\log y - y}\sqrt{2\pi}\cdots\,,$$

whence, differentiating the logarithm and neglecting the terms which contain negative powers of y, then the value is $= C + \log y$; hence, writing $y = \dfrac{1}{x}$, we obtain

$$\frac{1}{1+x} + \frac{1}{2(1+2x)} + \frac{1}{3(1+3x)} + \cdots = -C - \log x.$$

Writing herein $2x$ for x, and dividing by 2, we have

$$\frac{1}{2(1+2x)} + \frac{1}{4(1+4x)} + \cdots = -\tfrac{1}{2}C - \tfrac{1}{2}\log 2x = \tfrac{1}{2}(C - \log 2) - \tfrac{1}{2}\log x;$$

or, subtracting,

$$\frac{1}{1+x} + \frac{1}{3(1+3x)} + \frac{1}{5(1+5x)} + \cdots = -\tfrac{1}{2}(C + \log 2) - \tfrac{1}{2}\log x.$$

Hence, writing for x its value, $= \tfrac{1}{2}\pi a$, we have

$$S_1 = \tfrac{1}{4}(C + \log 2) - \tfrac{1}{4}\log\tfrac{1}{2}\pi a = \tfrac{1}{4}(C + 2\log 2 - \log\pi) - \tfrac{1}{4}\log a.$$

For the series S, we have (*Fundamenta Nova*, p. 103[6]) the formula

$$\frac{2K}{\pi} = \frac{1}{1+q^{-1}} - \frac{1}{3(1+q^{-3})} + \frac{1}{5(1+q^{-5})} - \cdots\,,$$

[6][* Jacobi's *Gesammelte Werke*, t. I, p. 159.]

or, putting herein $a = \dfrac{K'}{K}$, then $q = e^{-\frac{\pi K'}{K}} = e^{-\pi a}$, and thence

$$S = \frac{1}{1+e^{\pi a}} + \frac{1}{2(1+e^{2\pi a})} + \frac{1}{3(1+e^{3\pi a})} + \cdots = \log\sqrt{k'} + \frac{2K}{\pi}.$$

We have $q = e^{-\pi a}$, which is real, positive, and less than but indefinitely near to 1; hence also k is real, positive, and less than but indefinitely near to 1, say the value is $= 1-\beta$; thence $k' = \sqrt{(2\beta)}$, and $\log\sqrt{k'} = \frac{1}{4}\log 2\beta$; also $K' = \frac{1}{2}\pi$, and $K = \log\dfrac{4}{k'}$; therefore

$$a = \frac{\frac{1}{2}\pi}{\log\frac{4}{k'}} = \tfrac{1}{2}\pi \div \log\frac{2\sqrt{2}}{\sqrt{\beta}}, \quad \text{whence } \log\frac{2\sqrt{2}}{\sqrt{\beta}} = \frac{\pi}{2a},$$

which is the relation between a and β; and we thus have $\dfrac{2K}{\pi} = \dfrac{2}{\pi}\log\dfrac{2\sqrt{2}}{\sqrt{\beta}} = \dfrac{1}{a}$; and consequently $S = \frac{1}{4}\log 2\beta + \dfrac{1}{a}, = -\frac{1}{4}\log a$.

The two values thus are

$$S = -\tfrac{1}{4}\log a, \quad S_1 = \tfrac{1}{4}(C + 2\log 2 - \log\pi) - \tfrac{1}{4}\log a,$$

each depending on $\log a$, and having for this term the same coefficient $-\frac{1}{4}$; but there is in S_1 a constant term $\frac{1}{4}(C + 2\log 2 - \log\pi)$, where C is the constant $\cdot 577\ldots$.

It is easy to see why the series S is not reducible to S_1; however small a may be in the general term $\dfrac{1}{n(1+e^{n\pi a})}$, then taking n sufficiently large, not only $n\pi a$ is not indefinitely small, but it in fact becomes indefinitely large; the general term of the first series thus approximates to $\dfrac{1}{ne^{n\pi a}}$, or the terms more rapidly diminish somewhat than in a geometric series with the ratio $e^{-n\pi a}$ (a small positive value less than but very near to 1), whereas in the second series the general term approximates to $\dfrac{1}{\frac{n}{2}+(n+1)^2}$, or the convergence is ultimately that of the series $\dfrac{1}{n^2} + \dfrac{1}{(n+1)^2} + \cdots$.

902.

ON THE FOCALS OF A QUADRIC SURFACE.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 113–117.]

IN plane geometry, the focus of a curve is the node of the circumscribed line-system of the curve and the circular points at infinity; and so, in solid geometry, the focal of a surface or curve is the nodal line of the circumscribed developable of the surface or curve and the circle at infinity. And as in plane geometry the circular points at infinity may be regarded as an indefinitely thin conic, so in solid geometry the circle at infinity may be regarded as an indefinitely thin quadric surface.

In plane geometry, let it be proposed to find the circumscribed line-system (common tangents) of the two quadrics

$$\frac{x^2}{a}+\frac{y^2}{b}+\frac{z^2}{c}=0, \qquad \frac{x^2}{a'}+\frac{y^2}{b'}+\frac{z^2}{c'}=0;$$

if a common tangent be

$$\xi x+\eta y+\zeta z=0,$$

then we have

$$a\xi^2+b\eta^2+c\zeta^2=0,$$
$$a'\xi^2+b'\eta^2+c'\zeta^2=0,$$

here writing for shortness

$$f,\ g,\ h=bc'-b'c,\ ca'-c'a,\ ab'-a'b,$$

we have

$$\xi^2:\eta^2:\zeta^2=f:g:h;$$

and thence the tangent is

$$\sqrt{f}\,x+\sqrt{g}\,y+\sqrt{h}\,z=0,$$

viz. we have thus four tangents, and the rationalised form is of course

$$f^2x^4+g^2y^4+h^2z^4-2ghy^2z^2-2hfz^2x^2-2fgx^2y^2=0.$$

In connexion with the corresponding question in solid geometry, I obtain this equation in a different manner. We investigate the envelope of the line $\xi x+\eta y+\zeta z=0$, considering ξ,η,ζ as parameters connected by the foregoing two equations; by the ordinary process of indeterminate multipliers, we have

$$\xi x+\eta y+\zeta z=0,$$
$$a\xi^2+b\eta^2+c\zeta^2=0,$$
$$a'\xi^2+b'\eta^2+c'\zeta^2=0,$$

and thence, eliminating ξ,η,ζ, we obtain

$$\frac{x}{\lambda a+\mu a'}=\frac{y}{\lambda b+\mu b'}=\frac{z}{\lambda c+\mu c'},$$

$$\frac{ax^2}{(\lambda a+\mu a')^2}+\frac{by^2}{(\lambda b+\mu b')^2}+\frac{cz^2}{(\lambda c+\mu c')^2}=0,$$
$$\frac{a'x^2}{(\lambda a+\mu a')^2}+\frac{b'y^2}{(\lambda b+\mu b')^2}+\frac{c'z^2}{(\lambda c+\mu c')^2}=0,$$

equations equivalent to two equations, from which λ and μ are to be eliminated. The second and third equations are the derived functions $\frac{d}{d\lambda}, \frac{d}{d\mu}$ of the first equation in regard to λ, μ respectively; and hence, expressing the first equation in an integral form, the result is

$$\text{Disct.}\left[x^2(\lambda b + \mu b')(\lambda c + \mu c') + \text{\&c.}\right] = 0;$$

viz. this is

$$\{(bc' + b'c)x^2 + (ca' + c'a)y^2 + (ab' + a'b)z^2\}^2 - 4(bcx^2 + cay^2 + abz^2)(b'c'x^2 + c'a'y^2 + a'b'z^2) = 0,$$

or developing and reducing, we have the foregoing result, the coefficients entering through the combinations

$$f,\ g,\ h = bc' - b'c,\ ca' - c'a,\ ab' - a'b.$$

Writing $c = -1$, also $a' = b' = 1$, $c' = 0$: the equation of the first conic is

$$\frac{x^2}{a} + \frac{y^2}{b} - z^2 = 0,$$

and that of the second may be replaced by the two equations $x^2 + y^2 = 0$, $z = 0$, viz. these give the circular points at infinity: we have $f, g, h = 1, -1, a - b$, and the equation of the line-system is

$$x^4 + 2x^2y^2 + y^4 - 2h(x^2 - y^2)z^2 + h^2z^4 = 0.$$

If finally $z = 1$, then for the tangents from the circular points at infinity to the quadric

$$\frac{x^2}{a} + \frac{y^2}{b} - 1 = 0,$$

the equation is

$$x^4 + 2x^2y^2 + y^4 - 2h(x^2 - y^2) + h^2 = 0,$$

where, as before, $h = a - b$; the four tangents intersect in pairs in the two circular points at infinity, and in four other points which are the foci of the quadric.

Passing now to the problem in solid geometry, we consider the two quadric surfaces

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} + \frac{w^2}{d} = 0,$$

$$\frac{x^2}{a'} + \frac{y^2}{b'} + \frac{z^2}{c'} + \frac{w^2}{d'} = 0;$$

if a common tangent plane hereof be

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

then we have

$$a\xi^2 + b\eta^2 + c\zeta^2 + d\omega^2 = 0,$$

$$a'\xi^2 + b'\eta^2 + c'\zeta^2 + d'\omega^2 = 0,$$

and the circumscribed developable is the envelope of the plane, considering in the equation thereof ξ, η, ζ, ω as variable parameters connected by the last two equations. By what precedes, it is at once seen that the resulting equation is

$$\text{Disct.}\left[x^2(b\lambda + b'\mu)(c\lambda + c'\mu)(d\lambda + d'\mu) + \text{\&c.}\right] = 0,$$

viz. this is a quartic equation in (x^2, y^2, z^2, w^2), the coefficients $a, b, c, d, a', b', c', d'$, entering therein through the combinations

$$\mathfrak{a},\ \mathfrak{b},\ \mathfrak{c} = bc' - b'c,\ ca' - c'a,\ ab' - a'b;$$

$$\mathfrak{f},\ \mathfrak{g},\ \mathfrak{h} = ad' - a'd,\ bd' - b'd,\ cd' - c'd.$$

The developed result is given in my paper "On the developable surfaces which arise from two surfaces of the second order," *Camb. and Dub. Math. Jour.*, t. V. (1850), pp. 45–53, [84]. We require here, for the particular case, $d = -1$, $a' = b' = c' = 1$, $d' = 0$, viz. one of the surfaces is taken to be

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - w^2 = 0,$$

and the other is the circle at infinity $x^2 + y^2 + z^2 = 0$, $w = 0$; we thus have $\mathfrak{a}, \mathfrak{b}, \mathfrak{c} = 1, 1, 1$; $\mathfrak{f}, \mathfrak{g}, \mathfrak{h} = b - c, c - a, a - b$, or now using the italic letters f, g, h to signify these values, we have $f + g + h = 0$. The equation is

$$\begin{aligned}
&f^2x^8 + g^2y^8 + h^2z^8 + f^2g^2h^2w^8 \\
&\quad + 2f(f-h)g^2x^6y^2 + 2g(g-h)f^2x^2y^6 + 2h(h-f)g^2y^6z^2 + 2g^2(h-f)y^2z^6 \\
&\quad + 2h(h-g)f^2z^6x^2 + 2h^2(f-g)x^6z^2 \\
&\quad - 2f^2gh(g-h)x^6w^2 - 2fg^2h(h-f)y^6w^2 - 2fgh^2(f-g)z^6w^2 \\
&\quad + f^2(f^2 - 6gh)x^4w^4 + g^2(g^2 - 6hf)y^4w^4 + h^2(h^2 - 6fg)z^4w^4 \\
&\quad - 2gh(gh - 3f^2)w^4y^2z^2 + 2hf(hf - 3g^2)w^4z^2x^2 + 2fg(fg - 3h^2)w^4x^2y^2 \\
&\quad - 2f(hf - 3g^2)x^4y^2w^2 + 2g(fg - 3h^2)x^2y^4w^2 \\
&\quad - 2h(hf - 3g^2)z^4y^2w^2 - 2f(fg - 3h^2)z^2x^4w^2 \\
&\quad - 2(fg - 3h^2)x^4y^2z^2 - 2(hf - 3g^2)x^2y^4z^2 - 2(fg - 3h^2)x^2y^2z^4 \\
&\quad + \cdots = 0.
\end{aligned}$$

The equation may be written in the following four forms:

$$\begin{aligned}
x^2\Theta_1 + (gy^2 - hz^2 + ghw^2)^2\{y^4 + 2y^2z^2 + z^4 - 2f(y^2 - z^2)w^2 + f^2w^4\} &= 0, \\
y^2\Theta_2 + (hz^2 - fx^2 + hfw^2)^2\{z^4 + 2z^2x^2 + x^4 - 2g(z^2 - x^2)w^2 + g^2w^4\} &= 0, \\
z^2\Theta_3 + (fx^2 - gy^2 + fgw^2)^2\{x^4 + 2x^2y^2 + y^4 - 2h(x^2 - y^2)w^2 + h^2w^4\} &= 0, \\
w^2\Theta_4 + (x^2 + y^2 + z^2)^2 \times \{f^2x^4 + g^2y^4 + h^2z^4 - 2ghy^2z^2 - 2hfz^2x^2 - 2fgx^2y^2\} &= 0;
\end{aligned}$$

the last of these shows that the circle at infinity $x^2 + y^2 + z^2 = 0$, $w = 0$, is a nodal line on the surface; this, however, is not regarded as a focal. The other three show that we have also, as nodal lines, three conics, which are the focals of the given surface; viz. now writing $w = 1$, we have, for the quadric surface

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - 1 = 0,$$

the three focal conics

$$x = 0, \quad -\frac{y^2}{h} + \frac{z^2}{g} - 1 = 0,$$

$$y = 0, \quad \frac{x^2}{h} - \frac{z^2}{f} - 1 = 0,$$

$$z = 0, \quad -\frac{x^2}{g} + \frac{y^2}{f} - 1 = 0,$$

where, as before, $f, g, h = b - c, c - a, a - b$ respectively. If, as usual, a, b, c are positive and in order of decreasing magnitude, then for the ellipsoid

$$\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - 1 = 0,$$

we have the focal conics

$$\begin{aligned} x = 0, &\quad -\frac{y^2}{a-b} + \frac{z^2}{a-c} - 1 = 0, \\ y = 0, &\quad \frac{x^2}{a-b} - \frac{z^2}{b-c} - 1 = 0, \\ z = 0, &\quad \frac{x^2}{a-c} + \frac{y^2}{b-c} - 1 = 0; \end{aligned}$$

viz. these are an imaginary conic, the focal hyperbola, and the focal ellipse, in the coordinate planes $x = 0$, $y = 0$, $z = 0$ respectively.

903.

ON LATIN SQUARES.

[From the *Messenger of Mathematics*, vol. XIX. (1890), pp. 135–137.]

IF in each line of a square of n^2 compartments the same n letters $a, b, c, \ldots$ are arranged so that no letter occurs twice in the same column, we have what was termed by Euler "a Latin square." Supposing that in one of the lines the letters are arranged in the natural order $abcde \ldots$, then in the remaining lines there must be arrangements beginning with b, c, d, e, &c., respectively, and we may consider the case in which the bottom line has the arrangement $abcde \ldots$, and in the other lines, reckoning from the bottom one in order, the arrangements begin with b, c, d, e, &c., respectively: if the number of such squares be $= N$, then, obviously, the whole number of squares which can be formed with the same n arrangements is $= N\,[n]^n$.

Starting with the bottom line as above, then it is a well-known problem to determine the number of arrangements for the second line; viz. this number is

$$= 1 \cdot \underline{n} \ldots \left\{1 - \frac{1}{1} + \frac{1}{1 \cdot 2} - \cdots \pm \frac{1}{1 \cdot 2 \cdots n}\right\},$$

and if we assume, as above, that the second line begins with b, then the whole number of arrangements is this number divided by $(n-1)$, the quotient being of course integral. For instance, when $n = 5$, the number is $= 120 - 120 + 60 - 20 + 5 - 1, = 44$, which is divisible by 4, and the number of arrangements for the second line is thus $= 11$.

But the number of arrangements for the third line will be different according to the arrangement selected for the second line, and it is not easy to see how in general the whole number of arrangements for the third line is to be calculated, and the difficulty of course increases for the next following lines; it may be remarked that, when all the lines are filled up except the top-line, the top-line is completely determined.